Space-charge distortion of ionization profile monitor measurements in the CSNS rapid-cycling synchrotron

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Abstract
The ionization profile monitor (IPM) of the CSNS rapid-cycling synchrotron records the transverse proton distribution by collecting residual-gas electrons or ions. During their drift to the collector the secondaries are deflected by the bunch space-charge field, so that the detected r.m.s. width differs from the beam size. Particle tracking with Virtual-IPM 2.3.1 is used to quantify that error for the CSNS geometry and to assess the available mitigations. In electron mode a guiding field of 300 G reduces the relative expansion to the one-percent level for all beam families, intensities, cage voltages and orbit offsets examined; the design field of 0.1 T is conservative. In ion mode the same-sign kick always inflates the profile: at 100 kW a 10 mm beam is expanded by 92% (H_2^+, injection) and 104% (H_2^+, extraction). Raising the cage voltage from 5 to 30 kV reduces but does not remove the bias. The residual-gas species is identified in the time-of-flight spectrum; the true size is then recovered from one peak (H_2O^+ or N_2^+) by inverting a look-up of the expansion against bunch charge and cage voltage. The recommended operating point is electron collection at $B \gtrsim 300$ G.

Keywords
ionization profile monitor; space charge; CSNS RCS; electron collection; beam-size measurement

1 Introduction

An ionization profile monitor infers the transverse beam density from residual-gas ionization products collected on a position-sensitive detector [1, 4]. The CSNS rapid-cycling synchrotron (RCS) IPM [2, 3, 4, 5] uses a 220×231 mm cage biased at a design voltage of 25 kV ($E_y \simeq 108\,\mathrm{kV\,m^{-1}}$) and a magnet specified at 0.1 T (available to 0.2 T). Electrons are collected at the top electrode; the three ion species resolved in the time-of-flight spectrum—H_2^+, H_2O^+ and N_2^+—are collected at the bottom. Two proton bunches circulate. At 100 kW each carries $N = 7.8 \times 10^{12}$ particles (scaled linearly with beam power P). Injection is at 80 MeV ($\beta = 0.39$, $\sigma_t = 120$ ns, painted $\sigma_0 \simeq 25$ mm or a 10 mm core); extraction is at 1.6 GeV ($\beta = 0.93$, $\sigma_t = 20$ ns, $\sigma_0 \simeq 10$ mm).

While the secondaries drift they experience the bunch electric field and the Lorentz-boosted bunch magnetic field. The detected r.m.s. width σ_m is therefore not the beam width σ_0. The same mechanism has been analysed for magnetic electron IPMs by Vilsmeier, Sapinski and Storey [7] and for ion IPMs by Shiltsev [8] and by the ISIS group [9]. The question for CSNS is quantitative: how large is the error, how does it grow with bunch charge and shrink with beam size, and how efficiently do the available mitigations—guiding B, cage voltage, and an identified-species correction—remove it?

The calculation uses Virtual-IPM 2.3.1 [6] with uniform cage fields and 10^5 secondaries per case. Distortion is isolated by comparing bunch space charge on and off at fixed guiding E and B,

$$\Delta = \frac{\sigma_{\mathrm{SC}} - \sigma_{\mathrm{off}}}{\sigma_{\mathrm{off}}}\,. \tag{1}$$

A positive Δ is an inflated size; a negative Δ is compression. Microchannel-plate saturation, electromagnetic interference, secondary electrons from the ion trap and the measured cage map are not included. Those hardware effects, rather than space charge, are identified in Refs. [2, 3, 5] as the present electron-mode limit once a sufficient guiding field is applied.

2 Space-charge interaction with collected secondaries

Characteristic times fix the regime. At 25 kV an electron crosses the 115.5 mm half-gap in $\tau \simeq 3.5$ ns. The same path takes $\tau_2 \simeq 210$, 631 and 787 ns for H_2^+, H_2O^+ and N_2^+. The time to leave a 10 mm beam is $\tau_0 \simeq 74$, 221 and 275 ns. Injection bunches (120 ns) are longer than τ_0 for H_2^+; extraction bunches (20 ns) are shorter than every τ_0. The bunch spacing is 980 ns at injection and 409 ns at extraction.

Electrons are attracted to the proton bunch. A moderate kick focuses them towards the axis (compression). When the bunch field exceeds the cage field they cross the axis and the collected profile is wider than the beam (over-focusing). Peak transverse bunch fields on a 10 mm beam are approximately 29 kV m^{-1} (injection) and 73 kV m^{-1} (extraction) at 100 kW, versus 145 and 363 kV m^{-1} at 500 kW; the cage field is 108 kV m^{-1}. At 500 kW extraction electrons are trapped in the beam potential during the passage, the regime identified in Ref. [7] as requiring a magnetic field rather than an increase of V.

A guiding field converts the kick into cyclotron motion. If the impulse is delivered early in the flight, the displacement at the collector is proportional to $\sin(\omega_c\tau)/\omega_c$ and vanishes whenever $\omega_c\tau = n\pi$. Successive zeros of $\Delta(B)$ are therefore spaced by

$$\Delta B = \frac{\pi m_e}{e\tau} \propto \sqrt{V}\,. \tag{2}$$

The envelope of the oscillation falls as B increases. Magnetic mitigation is the decay of that envelope, not the choice of a particular zero. The same cyclotron-phase picture is the "one gyration" tuning rule used at GSI and J-PARC [10, 11].

Ions have the same sign as the beam, so the kick is always outward and Δ is always positive. In the impulsive limit (bunch length $\ll \tau_0$) the transverse impulse scales as N/σ_0^2 and the subsequent drift as $\tau_2 \propto 1/\sqrt{V}$, giving [8]

$$h - 1 \equiv \Delta \propto \frac{N}{V\,\sigma_0^2}\,, \tag{3}$$

with a mass dependence $\propto M^{-1/2}$. CSNS occupies a mixed regime: extraction is impulsive, whereas at injection H_2^+ leaves the beam while the 120 ns bunch is still passing. Magnetic confinement is an electron-mode instrument [7, 8] and is not treated further as an ion-mode correction.

3 Profile distortion and growth of the measured size

3.1 Electron collection without a guiding field

At $B = 0$ the collected electron width is not a reliable size. On the 10 mm injection beam at 100 kW, Δ changes from expansion to compression at 13 kV (+9% at 12 kV, ≈ 0 at 13 kV, -8% at 14 kV). At 500 kW the same voltage scan never changes sign: Δ remains positive and grows with V from +29% at 5 kV to +83% at 30 kV (Fig. 1). That is the trapping regime of Section 2: raising V does not un-trap a 500 kW bunch. The same intensity-dependent sign reversal is reported at GSI (electrons shrink without B at moderate intensity) [10] and at J-PARC (a factor-of-two shrink in electron mode) [11].

The map $\sigma_m(\sigma_0)$ is not monotonic at $B = 0$: a small beam may appear larger or smaller than a large one, according to whether the kick focuses or over-focuses. The Voitkiv H_2 double differential cross-section used by Virtual-IPM contributes a σ-independent quadrature smear of 2.9–3.2 mm ($\approx$ 2–2.5 eV per transverse axis), which is +40–48% at $\sigma_0 = 3$ mm and only +1% at 20 mm. One cyclotron period removes that smear (Section 4.1).

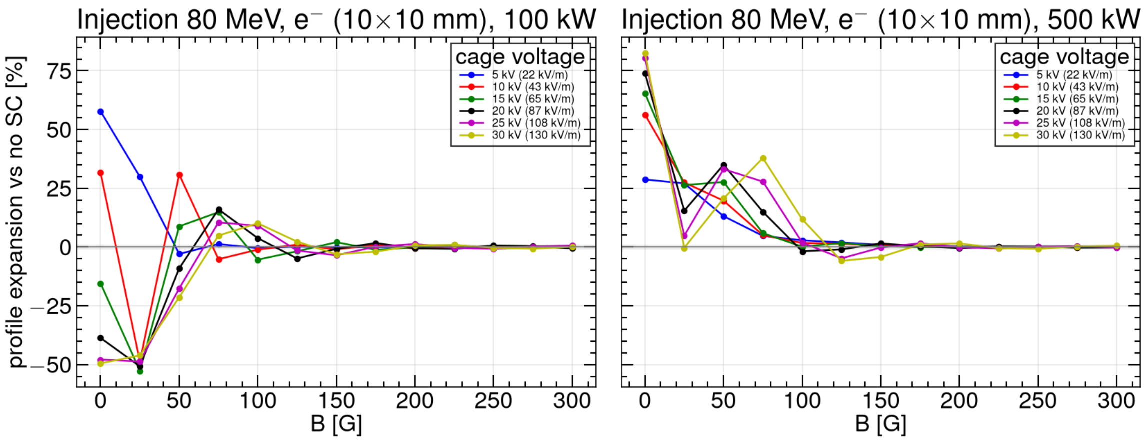

Fig. 1: Relative expansion Δ of the electron profile as a function of guiding field for cage voltages from 5 to 30 kV, on the 10 mm injection beam. At $B = 0$ the 100 kW family crosses zero; the 500 kW family does not.

Along the energy ramp at $B = 0$, a long bunch (120 ns) is compressed and a short bunch (20 ns) is inflated. Both errors become milder as β rises, because the electrons spend less time in the bunch: at 100 kW the 120 ns family goes from -46% (200 MeV) to -32% (1.6 GeV), and the 20 ns family from $+77\%$ (80 MeV) to $+38\%$ (1.2 GeV). Energy alone does not restore the profile.

3.2 Ion collection

Every ion case has $\Delta > 0$. Figure 2 shows σ_m versus σ_0 at the design 0.1 T and 100 kW: every species and both rings lie above the diagonal. Figure 3 shows the same map versus bunch charge: the expansion grows with P and falls with σ_0 as required by Eq. (3), and the observed width lies above $\sigma_m = \sigma_0$ at every power.

For H_2^+ at injection, $\sigma_0 = 10$ mm and 0.1 T, the expansion grows from $+82\%$ (18.2 mm) at 100 kW to $+175\%$, $+279\%$, $+390\%$ and $+422\%$ (52.1 mm) at 200, 300, 400 and 500 kW. Between 100 and 400 kW the growth is $\propto P^{1.12}$, consistent with the linear kick plus the second-order term of $\sigma_m^2 = \sigma_0^2 + \kappa N + \cdots$ [8]. At 500 kW the profile reaches the 110 mm half-cage and the exponent saturates.

At extraction, with ions born in time with the bunch, 10 mm, 100 kW and 25 kV, the expansions are $+104\%$ (H_2^+), $+41\%$ (H_2O^+) and $+40\%$ (N_2^+) at $B = 0$, in agreement with a reduced line-charge integration of the same equations to $\pm 1\%$ absolute (Fig. 4; model values $+102\%$, $+40\%$ and $+39\%$). The 20 ns bunch is impulsive, so H_2^+ is the most distorted species. At 500 kW the same family reaches $+408\%$, $+243\%$ and $+235\%$. Between $\sigma_0 = 5$ and 20 mm at 100 kW, $\Delta \propto \sigma_0^{-1.84\ldots-2.01}$, the impulsive σ_0^{-2} of Eq. (3) rather than the continuous-beam $\sigma_0^{-1.5}$. Aligned extraction H_2^+ widths at 0 G and 100 kW are listed in Table 1. A 3 mm extraction beam is already unusable as a size measurement at 100 kW. The most severe case remains injection at 3 mm and 500 kW, where $\sigma_m \approx 56$ mm ($+1700\%$) for all three species.

Table 1: Aligned extraction, H_2^+, 0 G, 100 kW, 25 kV: obtained width and expansion versus true size.

σ_0 (mm)	3	5	7	10	15	20
σ_m (mm)	38.2	26.8	22.4	20.4	21.7	24.9
Δ (%)	$+1172$	$+435$	$+219$	$+104$	$+44$	$+24$

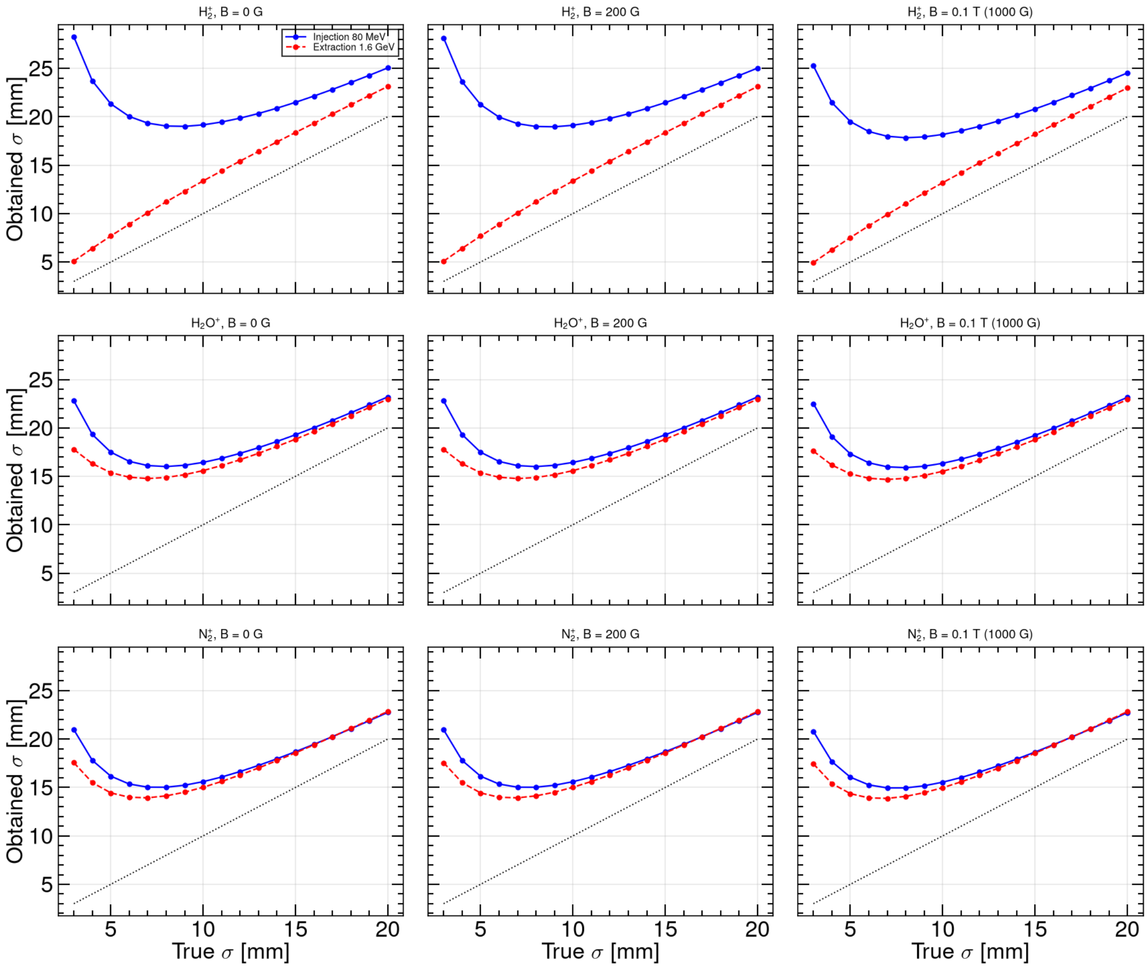

Fig. 2: Ion-mode obtained width σ_m versus true width σ_0 at $0.1\,\mathrm{T}$ and $100\,\mathrm{kW}$. The dashed line is $\sigma_m = \sigma_0$. Space charge inflates every point.

Species ordering follows the mixed regime. At injection, $10\,\mathrm{mm}$, $0\,\mathrm{G}$ and $100\,\mathrm{kW}$ the expansions are $+92\%$ (H_2^+), $+65\%$ (H_2O^+) and $+56\%$ (N_2^+): lighter ions expand more, but less than a pure $M^{-1/2}$ law because H_2^+ leaves during the $120\,\mathrm{ns}$ bunch. Painted injection ($25\,\mathrm{mm}$) is milder ($+17\%$, $+10\%$, $+9\%$). A second extraction bunch, still in the cage when the heavy ions arrive, adds a few points of extra width (single-bunch $+36\%$ and $+29\%$ versus three-bunch $+41\%$ and $+40\%$ for H_2O^+ and N_2^+); H_2^+ does not see that bunch ($+104\%$ either way). At commissioning power, aligned extraction H_2^+ ($10\,\mathrm{mm}$, $0\,\mathrm{G}$) is already $+19\%$ at $20\,\mathrm{kW}$ and $+82\%$ at $80\,\mathrm{kW}$. These magnitudes are of the same order as the operational ion-mode biases reported at J-PARC RCS (20–30% wider than the MWPM) [11] and at the Fermilab Booster [8].

4 Mitigation

4.1 Magnetic confinement of electrons

Figure 5 shows the electron expansion versus B for the three reference beams and 100–500 kW. The curves oscillate through focusing and over-kick. Painted injection recovers first; extraction and the $10\,\mathrm{mm}$ injection core still oscillate through 150–250 G. At 300 G every family has settled inside $\pm 1\%$ (Fig. 6).

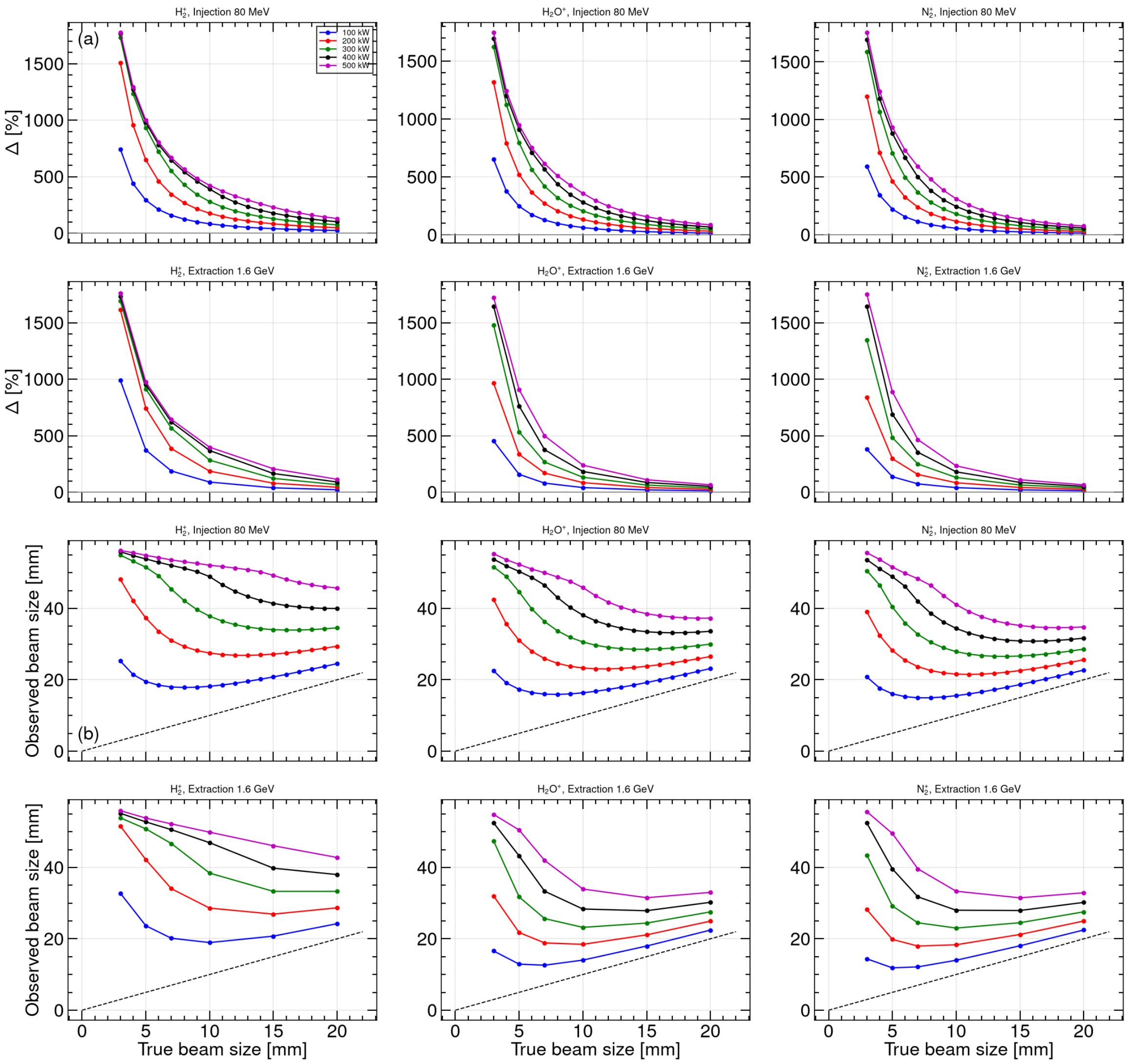

Fig. 3: (a) Ion-mode expansion Δ versus true size σ_0 at $0.1\,\mathrm{T}$ for 100–$500\,\mathrm{kW}$. Injection is the Block B size scan; extraction uses the aligned bunch timing (ion born with the bunch). (b) Observed width σ_m versus true width σ_0 for the same points. The dashed line is $\sigma_m = \sigma_0$. The $3\,\mathrm{mm}/500\,\mathrm{kW}$ family saturates on the cage wall ($\sigma_m \approx 56\,\mathrm{mm}$).

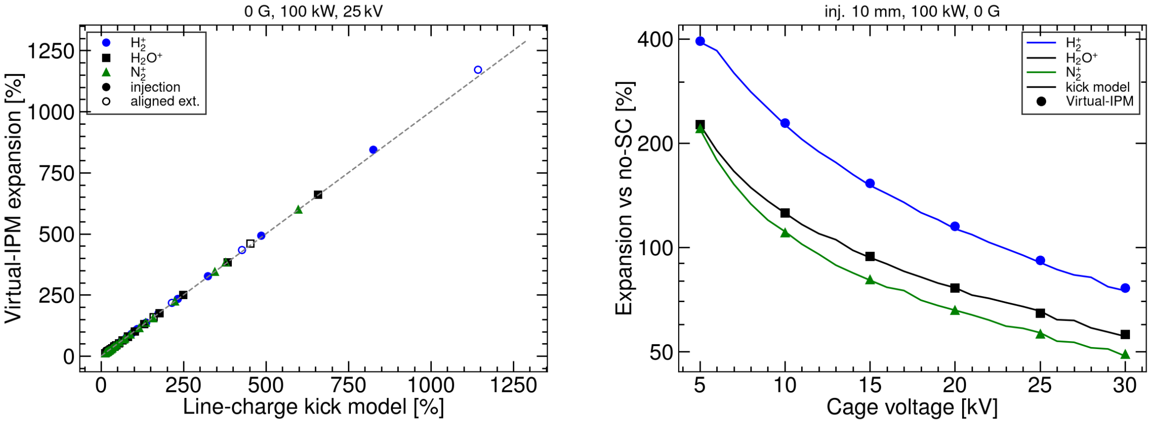

Fig. 4: Left: ion-mode expansion at 0 G, 100 kW and 25 kV, reduced kick model versus Virtual-IPM. Filled markers: injection $\sigma_0 = 3$–$20\,\mathrm{mm}$. Open markers: aligned extraction (3, 5, 7, 10, 15, 20 mm). Right: cage-voltage scan at 1 kV steps, injection 10 mm. Curves: kick model. Markers: Virtual-IPM. Agreement is $\pm 1\%$ absolute on the left and $\leq 4\%$ relative on the right.

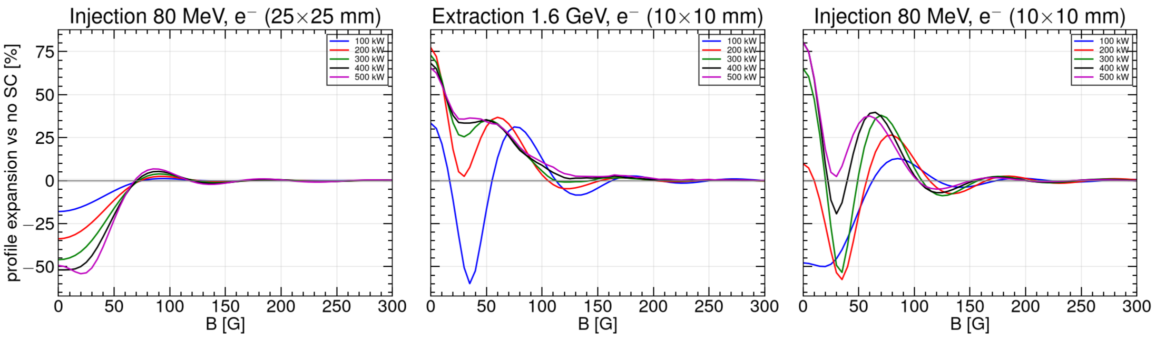

Fig. 5: Electron-mode expansion versus guiding field, 0–300 G, for painted injection (25 mm), extraction (10 mm) and the 10 mm injection core, at 100–500 kW. The grey band is $\pm 1\%$.

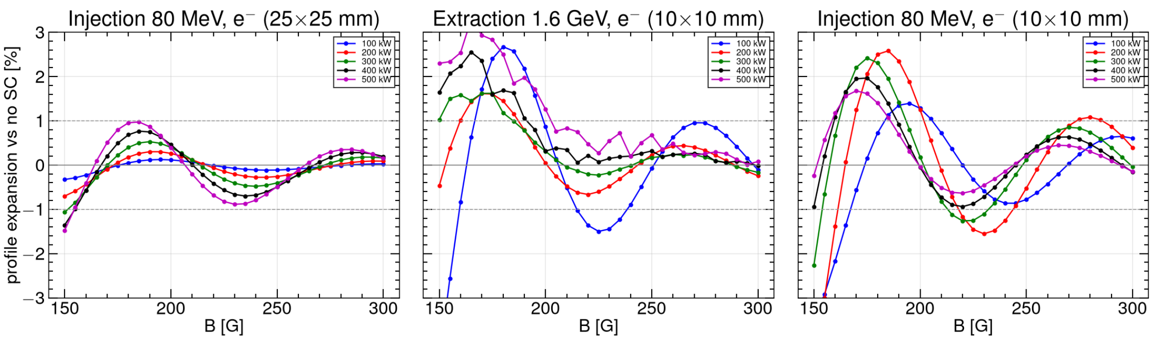

Fig. 6: As Fig. 5, 150–300 G, $\pm 3\%$ vertical scale. At 300 G every family lies inside $\pm 1\%$.

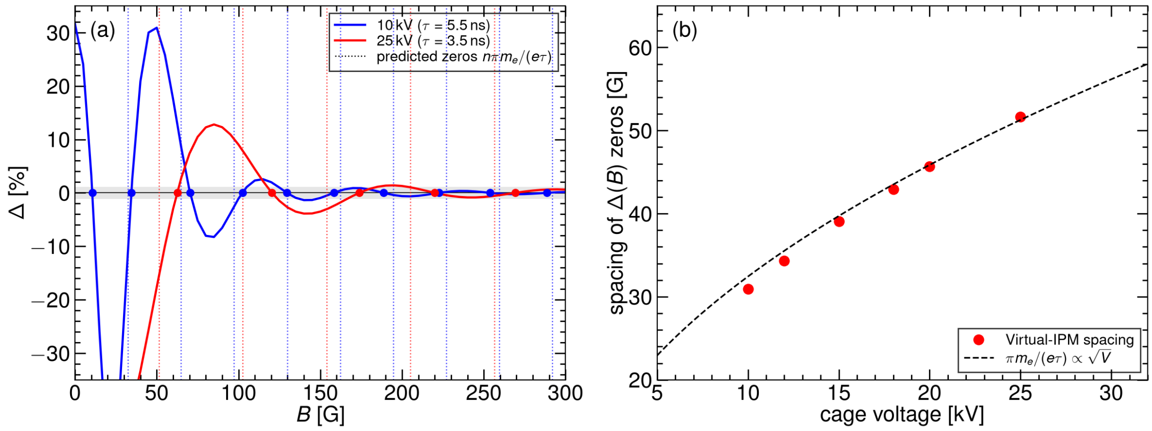

Fig. 7: (a) Relative expansion Δ versus guiding field at two cage voltages (100 kW, 10 mm injection). Dotted vertical lines (colour-matched to the curves) are the predicted zeros $B_n = n\pi m_e/(e\tau)$; circles are the simulated crossings. The grey band is $\pm 1\%$. The frame is $\pm 35\%$; the 25 kV point at $B = 0$ ($\Delta = -48\%$) remains off-scale. (b) Mean spacing of those zeros versus cage voltage at 1 kV steps, compared with $\pi m_e/(e\tau) \propto \sqrt{V}$ (no free parameter).

Figures 5 and 6 do not fall monotonically: $\Delta(B)$ oscillates through focusing ($\Delta < 0$) and overkick ($\Delta > 0$). Figure 7 identifies that oscillation as the cyclotron phase of a nearly impulsive space-charge kick.

An electron born on a 10 mm beam leaves the bunch in $\lesssim 1$ ns and then drifts the remaining $\tau \simeq 3.5$ ns (25 kV) in the uniform cage field. The transverse impulse Δv acquired inside the bunch is converted into cyclotron motion at $\omega_c = eB/m_e$. The displacement at the collector is $(\Delta v/\omega_c)\sin(\omega_c\tau)$ and therefore vanishes whenever the electron completes an integer number of half-turns, $\omega_c\tau = n\pi$ ($n = 1, 2, \ldots$). Those fields are exactly the equally spaced zeros $B_n = n\,\pi m_e/(e\tau)$ of Eq. (2). Because $\tau \propto V^{-1/2}$ in a uniform field, the spacing itself scales as $\sqrt{V}$. The amplitude of each successive lobe falls as $1/B$ (the $1/\omega_c$ prefactor). Magnetic mitigation is the decay of that envelope, not the choice of a particular zero: a zero can be moved by a few kilovolts or a change of orbit, whereas the envelope is a property of B.

Panel (a) is a 5 G-step scan at 100 kW and 10 mm injection, drawn at 10 and 25 kV, with vertical range $\pm 35\%$. Vertical dotted lines are the predicted zeros $B_n = n\pi m_e/(e\tau)$ for each voltage; filled circles mark the Virtual-IPM crossings. The 10 kV curve oscillates faster because the flight is longer ($\tau = 5.51$ ns versus 3.48 ns), so a given phase $n\pi$ is reached at a weaker B. The first 25 kV point, $\Delta = -48\%$ at $B = 0$, lies below the frame. At 25 kV the remaining lobes sit near 85, 140, 195, 240 and 295 G with amplitudes 13, 4, 1.4, 0.9 and 0.6%.

Panel (b) extracts the mean spacing between successive zeros on the 5–30 kV grid at 1 kV steps (Table 2 lists the 5 kV subset). The points follow the $\sqrt{V}$ curve to $\leq 5\%$ wherever the B scan is dense enough to interpolate the crossings, with no free parameter. That is the evidence that the oscillation in Figs. 5 and 6 is the cyclotron phase, and it is why 300 G—the first field at which the envelope itself lies inside $\pm 1\%$ at every power—is the operating point rather than any particular zero.

Table 3 lists the smallest field after which $|\Delta|$ remains below 1%, together with the residual at 300 G. Painted injection is restored first. The last curve to settle is the 10 mm injection core at 200 kW (290 G). A field of 300 G keeps every power at $|\Delta| \lesssim 1\%$.

The same 300 G point holds when the remaining axes are varied. From 200 G the obtained-versus-true plot lies on the diagonal except at 3 mm (Fig. 8). At 300 G, injection at 3 mm is still $+1\%$ (100 kW)

Table 2: Electron time of flight from the cage mid-plane and the spacing of $\Delta(B)$ zeros (dense B scan, injection 10 mm, 100 kW). The prediction is Eq. (2) with $\tau = \sqrt{2dm_e/(eE)}$ and $d = 115.5$ mm.

Cage voltage	ToF τ	ΔB predicted	ΔB simulated
10 kV	5.51 ns	32.4 G	30.9 G
12 kV	5.03 ns	35.5 G	34.3 G
15 kV	4.50 ns	39.7 G	39.1 G
18 kV	4.11 ns	43.5 G	42.9 G
20 kV	3.89 ns	45.9 G	45.7 G
25 kV	3.48 ns	51.3 G	51.7 G

Table 3: Smallest guiding field after which $|\Delta| \leq 1\%$ for the remainder of the 0–300 G grid. The residual at 300 G is given in parentheses.

Beam	100 kW	200 kW	300 kW	400 kW	500 kW
Inj. 25 mm	105 G (+0.03%)	115 G (+0.09%)	155 G (+0.16%)	155 G (+0.19%)	155 G (+0.15%)
Ext. 10 mm	240 G (−0.11%)	190 G (−0.24%)	185 G (−0.17%)	195 G (−0.02%)	205 G (+0.08%)
Inj. 10 mm	210 G (+0.61%)	290 G (+0.40%)	235 G (−0.04%)	190 G (−0.16%)	190 G (−0.15%)

to $+4\%$ (500 kW); 0.1 T places 3, 10 and 20 mm on the diagonal at every power ($|\Delta| \leq 0.05\%$). At 300 G and 10–25 kV, both 100 and 500 kW remain $\lesssim 1\%$. Higher V requires slightly more B at 100 kW (150 G at 10 kV to 210 G at 25 kV) because τ is shorter and the first lobes move out; 500 kW recovers earlier at every voltage. Along the energy ramp, 300 G and 100 kW give $|\Delta| \leq 0.5\%$ from 80 MeV to 1.6 GeV for both 120 ns and 20 ns bunches. At commissioning powers (20/50/80 kW) and 300 G the residuals are +0.00/+0.01/+0.02% (painted injection), +0.28/+0.56/+0.18% (extraction 10 mm) and +0.11/+0.29/+0.49% (injection 10 mm).

A few millimetres of vertical orbit offset change Δ by a few percent at $B = 0$ and by $\lesssim 1\%$ at 300 G; the 1% threshold moves by at most ~ 10 G. A horizontal offset is a translation of a uniform cage and does not change the width. The space-charge-induced centroid shift falls as $1/B$ (the $\mathbf{E} \times \mathbf{B}$ / polarisation drift of Ref. [7]): 0.9 mm at 100 G to 0.2 mm at 300 G on the extraction beam.

Thus $B = 300$ G reduces an electron-mode error of tens of percent (and the wrong sign at 100 kW and high V) to $\lesssim 1\%$. The design 0.1 T provides a factor-of-three margin and is the appropriate field for a 3 mm beam. The CSNS magnet (0.2 T) is not the limiting device. The SNS ring IPM was designed for 300 G at twenty-five times this bunch charge, with an estimated residual of about 7% [12]; a residual $\lesssim 1\%$ at CSNS is the expected scaling. The CERN PS beam-gas ionization monitor operates at 200 mT for LHC-type bunches [13], at the high-field end of the same scaling.

4.2 Cage voltage

For electrons at $B = 0$, voltage is not a size correction: it can reverse the sign of Δ at 100 kW and cannot reverse it at 500 kW (Section 3.1). Once $B \geq 300$ G the design 25 kV already lies inside 1%, so the voltage may be chosen for collection efficiency and detector operation.

For ions, a higher V shortens τ_2 and reduces the integrated kick, as at ISIS (broadening $\propto 1/E$) [9]. The relevant ion-mode scan is therefore $\Delta(V)$, not $\Delta(B)$: even at the design 0.1 T the ion cyclotron phase is $\omega_c\tau_2 \lesssim 1$ rad, so a guiding field is not a size correction.

Figure 9 is that voltage scan for the three time-of-flight-resolved species on the 10 mm injection beam at 100 kW and $B = 0$. Curves are the line-charge kick model at 1 kV steps from 5 to 30 kV; markers are Virtual-IPM on the same 1 kV grid. The model tracks every marker to $\leq 4\%$ relative (≤ 4 points absolute). Expansion remains positive at every voltage: from 5 to 30 kV, H_2^+ falls from +395% to +77%, H_2O^+ from +227% to +56%, and N_2^+ from +221% to +49% (Table 4). The Virtual-IPM points

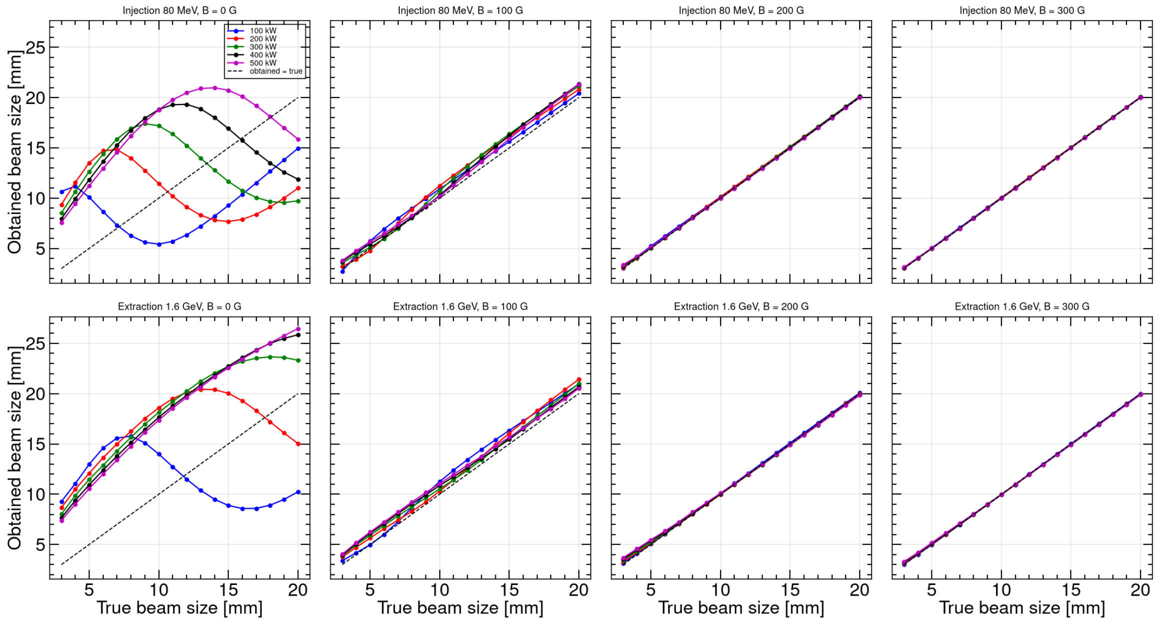

Fig. 8: Electron-mode obtained width versus true width at $B = 0$, 100, 200 and 300 G. From 200 G the points lie on the diagonal except at 3 mm.

scale as $V^{-0.92}$ (H_2^+), $V^{-0.77}$ (H_2O^+) and $V^{-0.83}$ (N_2^+). Aligned extraction H_2^+ (10 mm, 100 kW) falls from $+280\%$ at 5 kV to $+104\%$ at 25 kV and $+93\%$ at 30 kV. Extrapolating $V^{-0.92}$ from the 25 kV injection point, a 10% bias would require approximately 280 kV across the cage, which is not a practical path. ESS can operate an ion IPM at 300 kV m^{-1} because its bunches are weaker by about 5000 [14]; that choice does not transfer to CSNS.

Table 4: Ion-mode expansion versus cage voltage at $B = 0$ (injection 10 mm, 100 kW). Virtual-IPM on the 5 kV subset of the 1 kV scan.

V (kV)	5	10	15	20	25	30
H_2^+	+395	+229	+153	+115	+92	+77
H_2O^+	+227	+126	+94	+77	+65	+56
N_2^+	+221	+110	+81	+66	+56	+49

Voltage is therefore a useful lever on the ion bias (a factor of about five for H_2^+ from 5 to 30 kV) but not a substitute for correction or for electron-mode operation.

4.3 Closed-orbit offset

A few millimetres of closed-orbit error are routine in the RCS. The space-charge kick is centred on the beam, so a horizontal offset Δx is a translation of a uniform cage and must leave the measured width unchanged. A vertical offset Δy changes the drift length $d = 115.5\,\mathrm{mm} + \Delta y$ and therefore the flight time; that is the only first-order geometric lever.

Figure 10 is the electron check on the 10 mm beams at 100 kW. Panel (a) is the Fine D scan $\Delta y = \pm 10$ mm in 1 mm steps. At $B = 0$ extraction expands from $+25\%$ ($\Delta y = -10$ mm) to $+42\%$ ($\Delta y = +10$ mm), 0.85% per millimetre, because the electron stays longer in the bunch field when the beam is farther from the collector; injection at $B = 0$ is compressed ($\Delta \simeq -48\%$) and almost flat. At 300 G both beams lie inside the $\pm 1\%$ band over the whole ± 10 mm range ($|\Delta| \leq 0.65\%$). Panel (b) is the horizontal scan at that operating field in 1 mm steps from 0 to 10 mm: Δ is unchanged to the

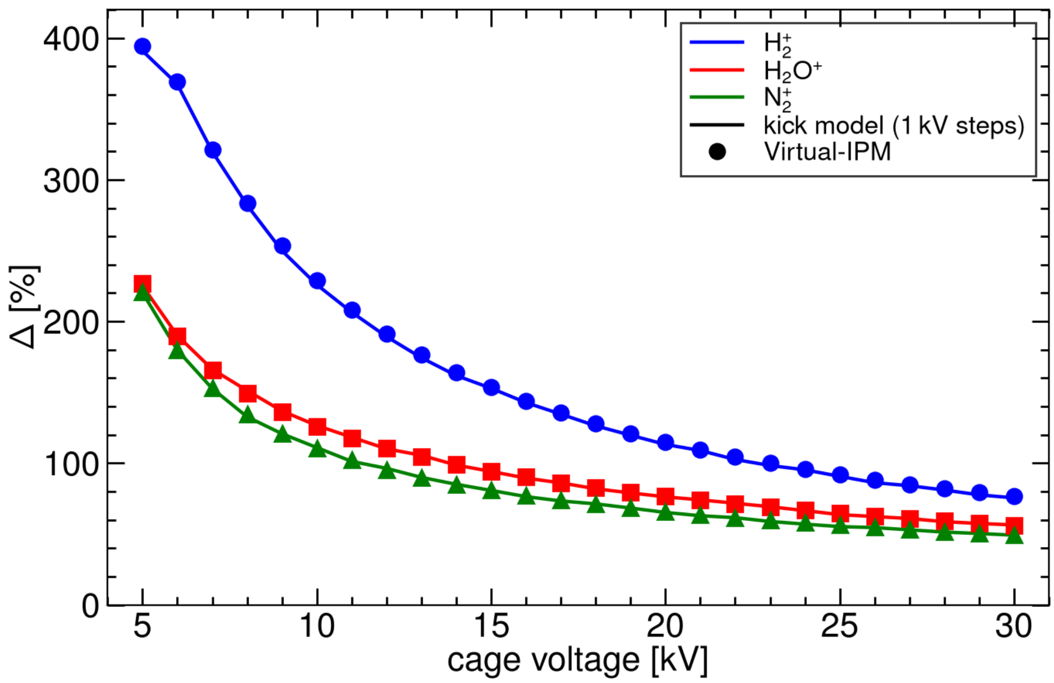

Fig. 9: Ion-mode expansion versus cage voltage for H_2^+, H_2O^+ and N_2^+ (injection $10\,\text{mm}$, $100\,\text{kW}$, $B = 0$). Curves: kick model at $1\,\text{kV}$ steps from 5 to $30\,\text{kV}$. Markers: Virtual-IPM on the same grid. Guiding B is not scanned.

line thickness (injection $+0.61\%$, extraction -0.11% at the centred point). The $\mathbf{E} \times \mathbf{B}$ centroid shift ($0.24\,\text{mm}$ on extraction at 300 G) is independent of Δx and is a fixed, correctable offset.

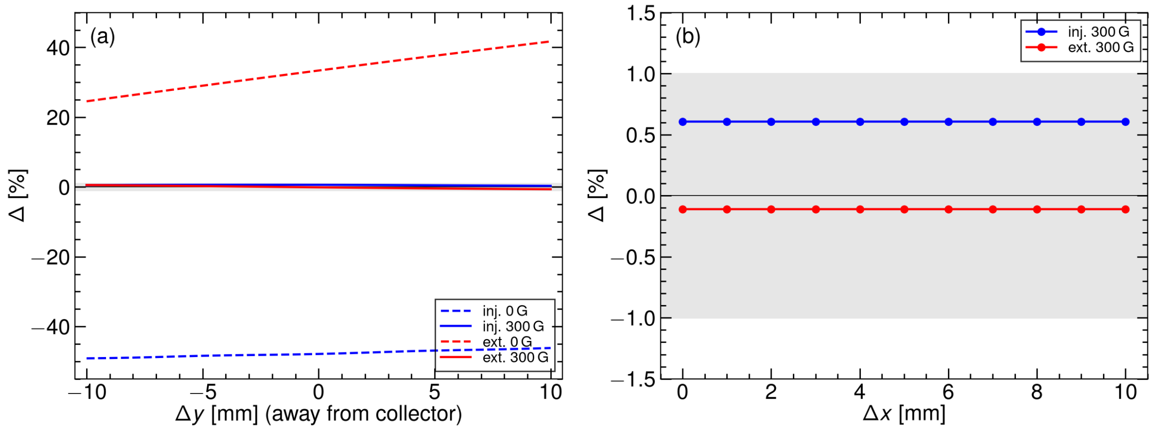

Fig. 10: Electron-mode closed-orbit offset, $10\,\text{mm}$ beams, $100\,\text{kW}$. (a) Δ versus Δy (Fine D, $1\,\text{mm}$ steps). Dashed: $B = 0$. Solid: 300 G. The grey band is $\pm 1\%$. (b) Δ versus Δx at 300 G (0–$10\,\text{mm}$, $1\,\text{mm}$ steps).

Figure 11 is the ion check at $B = 0$ (a guiding field is not an ion-mode lever). Panel (a) plots Δ versus Δy for the three species. Filled markers are injection; open markers are aligned extraction. Lines are $\sigma_m^2 - \sigma_0^2 \propto d$, anchored at the centred Virtual-IPM point, with no free parameter. Injection H_2^+ is $+89.2\%$, $+92.0\%$ and $+94.7\%$ at $\Delta y = -5$, 0 and $+5\,\text{mm}$; the prediction is $+88.9\%$ and $+95.0\%$. Aligned extraction H_2^+ is $+101\%$, $+104\%$ and $+107\%$ on the same Δy grid. The $1\,\text{mm}$ scan from -5 to

$+5$ mm stays on that line. Every species and both beams agree with the drift-length law to 0.5% absolute. Panel (b): the 1 mm Δx scan from 0 to 10 mm leaves every width unchanged to 0.1%, as required by a kick centred on the beam and as stated for ISIS within about 10 mm of the axis [9].

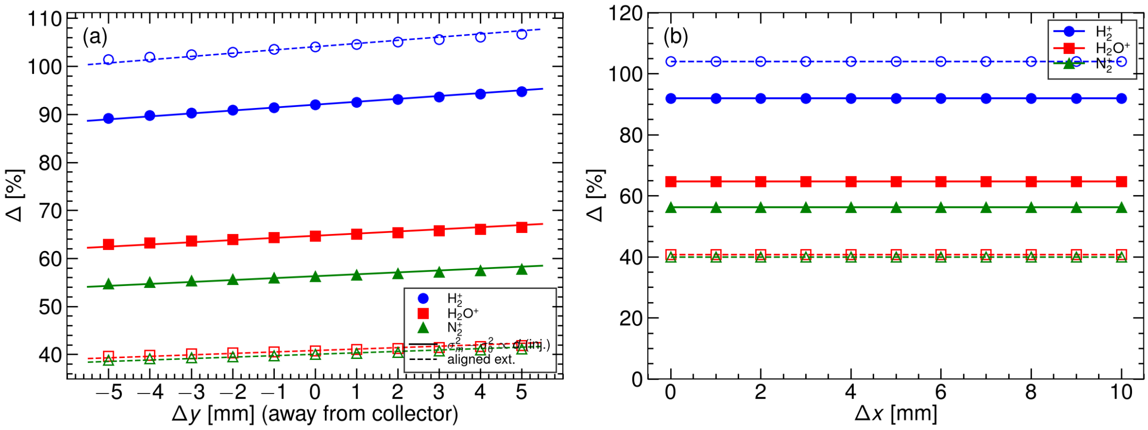

Fig. 11: Ion-mode closed-orbit offset at $B = 0$, 10 mm, 100 kW, 25 kV. (a) Δ versus Δy (-5 to $+5$ mm, 1 mm steps). Filled: injection. Open: aligned extraction. Lines: $\sigma_m^2 - \sigma_0^2 \propto d$. (b) Δ versus Δx (0–10 mm, 1 mm steps); the width is translation-invariant. Guiding B is not scanned.

At 500 kW an uncorrected injection H_2^+ profile ($\sigma_m \approx 52$ mm) is clipped by the 110 mm half-cage. A 10 mm horizontal offset then pulls the centroid by 4 mm. That is an aperture effect, not a failure of the electron-mode operating point and not a space-charge shift of the ion image.

4.4 Inversion of ion-mode widths

The ion bias is large, smooth, and reproduced by the kick model (Fig. 4), so σ_0 can be recovered from σ_m without changing the hardware. The residual-gas species is identified from the time-of-flight spectrum. The working peak is then a single identified species—H_2O^+ or N_2^+, which are less impulsive than H_2^+ (Section 2)—and σ_0 is read from that species' look-up $\sigma_m(\sigma_0, N, V)$ at $B = 0$, 25 kV (Virtual-IPM Block B / I2). On the Fermilab Booster the same idea is a closed-form h started at $\sigma_0 = \sigma_m$ and quoted at 5–10% [8]. CSNS substitutes a tabulated h for the mixed regime and an identified ToF peak for an unresolved mixture.

Figure 12(a) is the collected width versus the true size for the three species at injection and 100 kW. Each curve has a minimum (H_2^+: $\sigma_m = 19.01$ mm at $\sigma_0 = 9$ mm; H_2O^+: 16.01 mm at 8 mm; N_2^+: 15.02 mm at 8 mm). A width above that minimum has two roots, so the interpolant is taken on the physical branch $\sigma_0 \geq \sigma_0^*(P)$ (the location of the minimum). Injection cores and painted beams sit on that branch at 100 kW. The closed-form Shiltsev law $\Delta \propto N/(V\sigma_0^2)$ has only the large root of $\sigma_m = \sigma_0 + \kappa/\sigma_0$ and cannot represent the minimum.

H_2^+ is not the working peak. At the 10 mm design core its slope is only $\mathrm{d}\sigma_m/\mathrm{d}\sigma_0 \simeq 0.22$, the two roots (7.6 and 10 mm) sit next to the fold, and a leave-one-out invert on $\sigma_0 =$ 8–20 mm has a maximum error of 15%. H_2O^+ and N_2^+ are steeper at 10 mm (0.36 and 0.42) and better conditioned.

The operational sequence is: identify the H_2O^+ or N_2^+ ToF peak; read that σ_m and the DCCT bunch charge N; interpolate $\sigma_m(\sigma_0, N, V)$ from the Block B / I2 table of the identified species on $\sigma_0 \geq \sigma_0^*(P)$; return σ_0. Cage voltage and a BPM Δy enter as parameters of the same table ($\Delta \propto 1/V$ and $\sigma_m^2 - \sigma_0^2 \propto d$, Sections 4.2 and 4.3). The extraction column must be the aligned generation (Fig. 14; H_2^+ $+104\%$ at 10 mm, 100 kW), not the v1 as-run $+33.5\%$.

Figure 13 applies that invert to the Virtual-IPM points, leaving the tested σ_0 out of the interpolant.

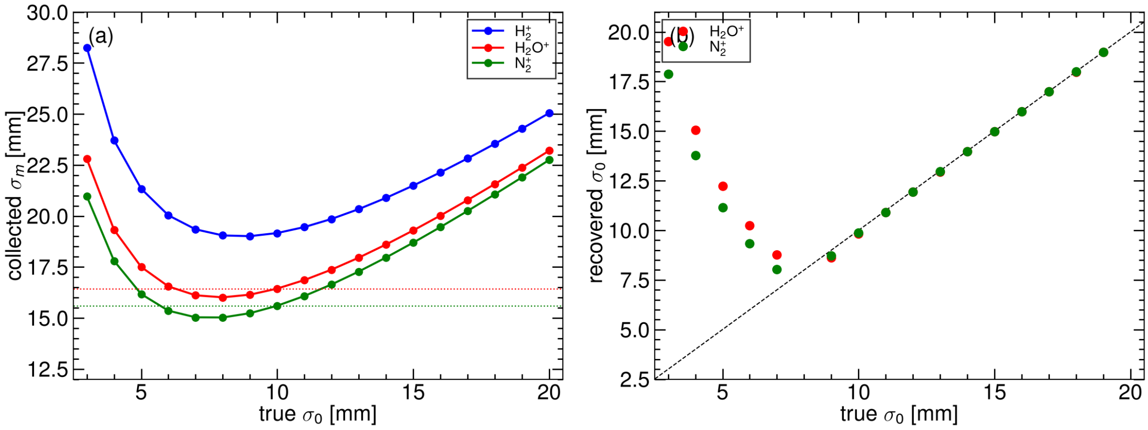

Fig. 12: Injection, 100 kW, 25 kV, $B = 0$. (a) Collected σ_m versus true σ_0. Dotted lines are the H_2O^+ and N_2^+ widths of a 10 mm beam (16.44 and 15.60 mm). (b) Leave-one-out invert of the identified H_2O^+ or N_2^+ table on $\sigma_0 \geq \sigma_0^*$. Empty markers at $\sigma_0 < 8$ mm have no root on that branch.

Panel (a), injection 100 kW: the uncorrected N_2^+ σ_m lies far above the diagonal; the identified H_2O^+ and N_2^+ tables recover 8–20 mm to a median 0.25% and 0.21% (maxima 4.3% and 3.0% at the fold). The 10 mm design point returns 9.83 mm (H_2O^+) and 9.86 mm (N_2^+). A 5 mm beam has no root on $\sigma_0 \geq 8$ mm and is not an RCS operating size at this intensity. Panel (b): the same identified-species invert at 200 kW stays inside a few percent wherever the profile is inside the cage (median 2.0% and 1.6%); the fold has moved out to 12 and 11 mm, so a 10 mm core is then the small root and must be selected as the core, not the paint, hypothesis. At 300 kW N_2^+ is still usable (median 2.4%, nine sizes). At 400 and 500 kW the heavies clip ($\sigma_m \simeq 31$–47 mm, detected fraction < 0.995) and no table recovers σ_0.

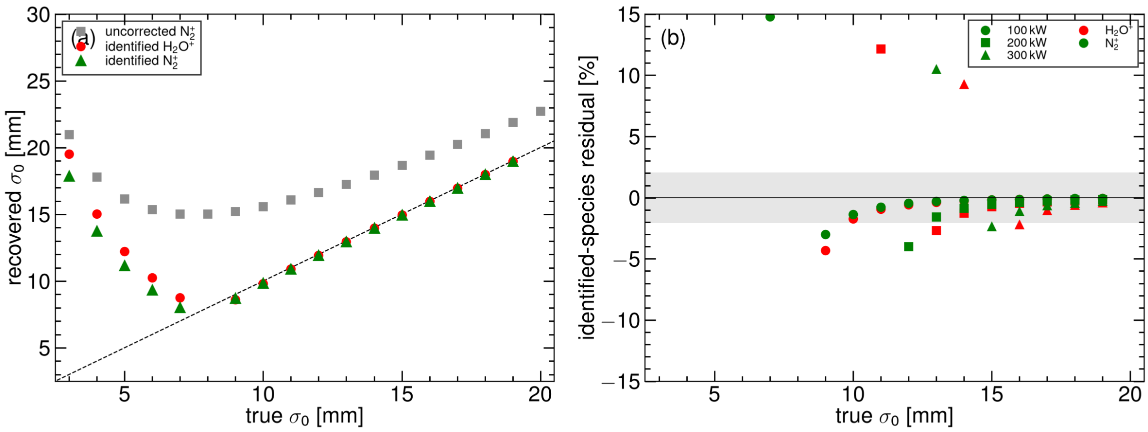

Fig. 13: (a) Injection 100 kW: uncorrected N_2^+ σ_m and the identified H_2O^+ and N_2^+ inverts versus true σ_0 (leave-one-out). (b) Identified-species residual at 100–300 kW. Red: H_2O^+. Green: N_2^+. The grey band is $\pm 2\%$. Open markers are wall-clipped families.

Aligned extraction at 100 kW has the same fold (H_2O^+ minimum 12.7 mm at 7 mm; N_2^+ 11.9 mm at 5 mm). The 10 mm design size is already on the rising branch ($\sigma_m = 14.1$ and 14.0 mm). The I1 size grid is too coarse (3, 5, 7, 10, 15, 20, 25 mm) for a leave-one-out invert at the 0.2% injection level; a production table should be filled at 1 mm in σ_0, as Block B already is at injection. Field-cage nonuniformity—J-PARC reported a factor-of-two shrink from the external map alone [11]—is not in the table and will have to be folded in once the CSNS measured map is available.

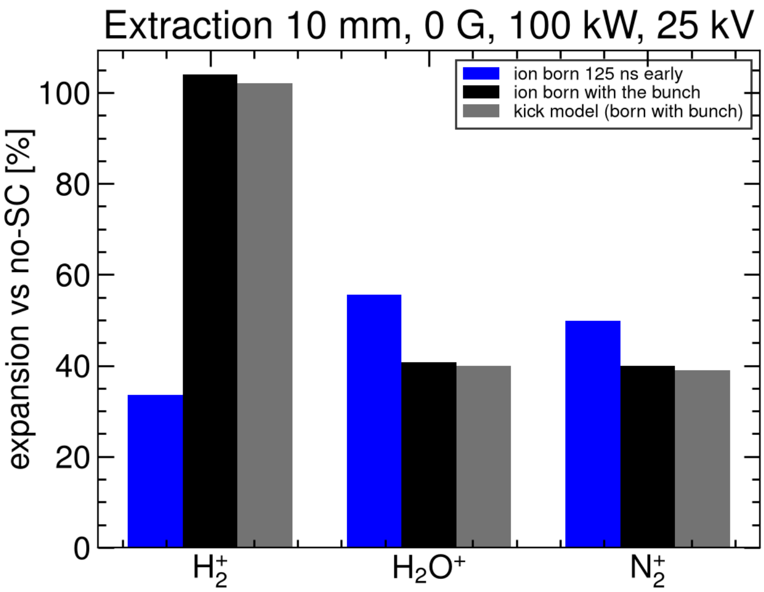

Fig. 14: Extraction, $10\,\mathrm{mm}$, $0\,\mathrm{G}$, $100\,\mathrm{kW}$, $25\,\mathrm{kV}$. Virtual-IPM with the ion born in time with the bunch matches the kick model. This is the extraction column of the inversion table.

4.5 Summary

Table 5 collects the efficiencies. Magnetic confinement at $300\,\mathrm{G}$ is the only mitigation that returns a faithful electron size. Cage voltage reduces the ion bias but does not bring an uncorrected $10\,\mathrm{mm}$, $100\,\mathrm{kW}$ profile to the 10% level. An identified $\mathrm{H_2O^+}$ or $\mathrm{N_2^+}$ invert does, at 100–300 kW, provided the ions stay inside the cage.

Table 5: Efficiency of the mitigations examined. Residuals are quoted for the families of Sections 3 and 4.

Mitigation	Electron mode	Ion mode
$B = 300\,\mathrm{G}$	tens of percent $\to \lesssim 1\%$ (all P, V, energy; $\sigma \geq 4\,\mathrm{mm}$)	—
Design $B = 0.1\,\mathrm{T}$	$3\,\mathrm{mm}$ on the diagonal; residual $\leq 0.05\%$	—
Raise V ($5 \to 30\,\mathrm{kV}$)	not a size fix at $B = 0$; unused once $B \geq 300\,\mathrm{G}$	$+395\% \to +77\%$ ($\mathrm{H_2^+}$, inj. $10\,\mathrm{mm}$, $100\,\mathrm{kW}$, $B = 0$); 10% would require $\sim 280\,\mathrm{kV}$
Orbit $\lesssim 10\,\mathrm{mm}$	$300\,\mathrm{G}$ still $\lesssim 1\%$	Δx: none; Δy: a few percent
Invert $\sigma_m = \sigma_0\, h(\sigma_0, N, V)$	unnecessary at $\geq 300\,\mathrm{G}$	identify $\mathrm{H_2O^+}$ or $\mathrm{N_2^+}$; that peak recovers σ_0 to 0.2% (inj. $100\,\mathrm{kW}$, 8–$20\,\mathrm{mm}$); $\mathrm{H_2^+}$ is poorly conditioned at $10\,\mathrm{mm}$; fails at the wall ($P \gtrsim 400\,\mathrm{kW}$)

5 Conclusions

Space-charge distortion of the CSNS RCS IPM has been quantified with uniform-field tracking and checked against a reduced kick model and against the published electron-mode and ion-mode scalings.

Electrons experience an attractive kick that can focus or over-focus. The resulting $\Delta(B)$ is a cyclotron-phase oscillation whose envelope falls below 1% at $300\,\mathrm{G}$ for every beam, power, voltage and offset examined, and along the energy ramp. Size growth with charge, and the non-monotonic $\sigma_m(\sigma_0)$ at $B = 0$, both disappear at that field. The design $0.1\,\mathrm{T}$ is a margin, except for the smallest ($3\,\mathrm{mm}$) beams.

Ions experience a repulsive kick and are always collected with $\sigma_m > \sigma_0$. The inflation scales as N/σ_0^2 (impulsive) and as approximately $1/V$, and matches the kick model to 1%. Raising the cage voltage cannot bring a $10\,\mathrm{mm}$, $100\,\mathrm{kW}$ ion profile to the 10% level. A look-up inversion of one identified species (H_2O^+ or N_2^+) recovers σ_0 to 0.2% at $100\,\mathrm{kW}$ on $\sigma_0 = 8$–$20\,\mathrm{mm}$, provided the profile has not reached the wall. H_2^+ is not used for the size.

The recommended operating point is therefore electron collection at $B \gtrsim 300\,\mathrm{G}$. Ion time-of-flight identifies the residual gas; the beam profile is then taken from the H_2O^+ or N_2^+ peak after inversion. Once space charge is removed, the remaining electron-mode systematic is hardware: microchannel-plate saturation after $\sim 200\,\mu\mathrm{s}$, electromagnetic interference, secondary electrons, and the real cage map [2, 3, 5].

Acknowledgements

The hardware parameters follow Ref. [4]. Tracking was performed with Virtual-IPM 2.3.1 [6]. This note is prepared for internal review.

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